Sahiel Bose

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Education

University of California, Los Angeles — **Los Angeles, CA**
B.S. in Mathematics of Computation

Experience

Optivia — **Dublin, CA**
Co-Founder — *March 2025 – Present*

- Lead architecture of a multi-agent pipeline that converts a natural-language prompt into a dependency-graphed fleet of specialized autonomous coding sub-agents.
- Designed the underlying math – a five-signal complexity scoring formula, critical-path (CPM) scheduling, and a per-agent reflection layer to prevent hallucination propagation (patent pending).
- Driving Optivia's long-term vision: enabling highly technical, long-running enterprise projects to be executed by personalized, cost-efficient agents built for better accuracy, lower cost, and detailed functionality.

ThinkNeuro LLC — **Dublin, CA**
Neurotechnology & Software Engineering Intern — *November 2024 – Present*

- Leading machine learning integration for a patent-pending wearable neurotechnology device aimed at real-time opioid overdose detection, owning the signal processing and modeling stack end-to-end.
- Spearheaded development of GRU, LSTM, and logistic regression classifiers on a 721K-sample physiological dataset, using subject-level splits and class-balanced loss to reduce false detections and lift real-time accuracy.
- Co-authoring a research paper and literature review on the biphasic overdose physiological signature, and leading development of a React Native mobile companion application for real-time alerts.

UC Santa Cruz AIEA Lab — **Santa Cruz, CA**
AI Research Intern — *October 2024 – April 2026*

- Researching logical reasoning in large language models (LLMs) and contributing to the ProSLM project, which enhances LLM problem-solving through structured prompting and verification.
- Designing structured exercises and chain-of-thought scaffolds, and exploring algorithmic techniques like self-consistency to refine and optimize LLM reasoning processes.

UC Riverside Research Lab — **Riverside, CA**
Machine Learning Research Intern — *May 2025 – January 2026*

- Assisted a PhD mentor on a Graph Neural Network (GNN) project detecting traffic congestion hotspots in heterogeneous road-network graphs (intersection + event nodes) via GraphSAGE/GraphSAINT sampling.
- Supported data preprocessing, model training, and hyperparameter tuning while evaluating modern sampling strategies (GRAPES, HMSG) to improve scalability and predictive insights for urban planning.

Projects

HealthFlow - Apify x Scalekit | *Multi-Agent AI, Next.js* — **May 2026**

- Won **1st Place** at the Scalekit x Apify Hackathon for an end-to-end multi-agent healthcare AI pipeline.
- Built a 9-agent TypeScript/Next.js system that converts paramedic voice input into structured FHIR R4 data, runs AI-driven differential diagnoses, and commits physician-approved orders to the EHR in under 60 seconds.
- Integrated Apify-powered drug interaction and allergy screening with Scalekit authentication and a SHA-256 immutable audit trail for end-to-end clinical safety and compliance.

Opioid Technology Device | *ML, Embedded Deployment* — **November 2024 – Present**

- Built and benchmarked three ML classifiers (logistic regression, GRU, LSTM) on a 721K-sample physiological dataset, with the GRU selected for on-device inference via TFLite INT8 quantization.
- Engineered domain-specific features capturing the biphasic overdose signature and conducted FMEA-style risk analysis to guide hardware-software integration (patent application in progress).

UC Riverside HotSpot Detection with GNNs | *Heterogeneous GNNs* — **May 2025 – January 2026**

- Developed heterogeneous GNN models over spatial road-network graphs with intersection and event node types to identify traffic congestion hotspots.
- Benchmarked sampling-based architectures including GraphSAGE, GraphSAINT, and GRAPES on smaller datasets to validate scalability for large-scale transportation graphs.

ACSEF Project: Sign Language Recognition System | *CNNs, LSTMs* — **December 2024 – March 2025**

- Earned **2nd Place** at the Alameda County Science & Engineering Fair (ACSEF).
- Co-led an AI-powered ASL recognition system using MobileNetV2 transfer learning and a bidirectional LSTM, achieving 95% validation accuracy and deployed on Raspberry Pi for real-time inference.

Technical Skills

AI Techniques: Agentic Workflows, Multi-Agent Systems, Transfer Learning, Reinforcement Learning, Model Optimization, Logical Reasoning, Sequence Modeling
Frameworks & Libraries: TensorFlow, PyTorch, Keras, React Native, Next.js, MCP
AI Models & Applications: Graph Neural Networks, Large Language Models, Computer Vision, AI Agents